

Aaron Chou

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EDUCATION

- Carnegie Mellon University**, *Master of Science in Robotic Systems Development* Pittsburgh, PA
• **Coursework:** Introduction to Robot Learning, Robot Autonomy, Robot Mobility May 2028
• **MRSD Capstone:** Developing **video-based learning** methods for humanoid manipulation tasks
- National Taiwan University**, *Bachelor of Science in Mechanical Engineering*, GPA: 3.80/4.3 Taipei, Taiwan
• **Coursework:** Automatic Control, Digital Control System, Kinematics, Dynamics, Computer Programming Jan. 2023

PUBLICATIONS

- [1] **Y.L. Chou**, L.C. Wang, Y.S. Luo, Y.T. Liu, J.L. Li. *VL-N3RD-Bench: Benchmarking Vision-Language Navigation with 3D Gaussian Splatting Reconstruction for Deployment*, ICRA Workshop: VLA Pipelines for Real Robots, 2026.
- [2] Y.S. Luo, L.C. Wang, H. Mandala, **Y.L. Chou**, G.H.G. Christmann, C.Y. Lee, W.C. Chen, et al. *Feasibility-Guided Planning over Multi-Specialized Locomotion Policies*, IEEE Int. Conf. Robot. Autom. (ICRA), 2026.

EXPERIENCE

- Stealth-Mode Startup** Mountain View, CA
Robotics Research Intern — UMI, Diffusion Policy May 2026 – Aug. 2026
- Inventec Corporation**, AI Center, Robotics Taipei, Taiwan
Robotics Research Engineer — Quadruped Robots Mar. 2025 – May 2026
- **Vision-Language Navigation (VLN):** Integrated and evaluated a pre-trained VLA model for real-world navigation, and built an **IsaacLab** benchmark across scenes reconstructed with **3D Gaussian Splatting** (3DGS)
 - **3DGS Benchmarking:** Benchmarked 4 pipelines on success rate, image quality, runtime, and GPU memory; published in [1]
 - **Learning-Based Control:** Developed low-level control infrastructure with LCM communication, custom motor interfaces, calibration routines, and IMU integration for quadruped platforms with no official API, deploying **IsaacGym**-trained policies
 - **Terrain-Aware Navigation:** Trained a 15 cm gap-traversal policy, implemented a feasibility-guided planning framework for hybrid terrain skill selection, conducted real-world experiments, and contributed to an ICRA 2026 publication [2]
 - **SLAM Optimization:** Improved single-LiDAR SLAM by **dual-LiDAR fusion** in ICRA 2025 QRC, achieving a **65%** reduction in mapping-to-navigation time and enabling rapid terrain reconstruction in dynamic environments
- Flytech Technology Co. Ltd.** Taipei, Taiwan
Robotics Engineer — Autonomous Mobile Robots Mar. 2024 – Jan. 2025
- Engineered a **precise docking system** by integrating motion control, localization, and a customized rack tracker, achieving navigation accuracy of **±1–1.5 cm** to minimize re-docking attempts in production deployments
 - Developed and maintained **10+ ROS packages** and delivered a patrol demo showcased at Computex 2024
- Chien Kuo High School**, Robotics team, FRC#8020 CyberpunkK Taipei, Taiwan
Youth Mentor — Mechanical Design Feb. 2022 – Jul. 2023
- Mentored 30+ students for robot design, CAD/CAM, CNC operation, and competition-ready robot testing

SKILLS

Robotics: IsaacSim/Lab, MuJoCo, Imitation/Reinforcement Learning, VLM/VLA, ROS/ROS2, 3D Reconstruction, Sensor Fusion
Hardware & Software: UMI, Flexiv Rizon, Unitree G1/Go1/A1, Livox LiDAR, RealSense, IMU; Docker, Git, Blender, Onshape
Programming: Python, C++, Shell Scripting (Bash), LaTeX

HONORS & COMPETITIONS

- 2025 ICRA Quadruped Robot Challenge (QRC) — Participation Certificate, Autonomous** Atlanta, GA
- 2022 FIRST Robotics Competition (FRC) Sacramento Regional, Finalist (Team Mentor)** Sacramento, CA
- 2020 Dean's List Award (top 5% of class)** Taipei, Taiwan